

Industrial Internship Report on “Python-Based Application Development”

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

The internship focused on practical Python application development. Four projects were completed: URL Shortener, Secure Password Manager, Quiz Game and File Organizer.

The work provided practical exposure to web development, GUI development, databases, encryption, file handling, debugging and testing.

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1 Preface

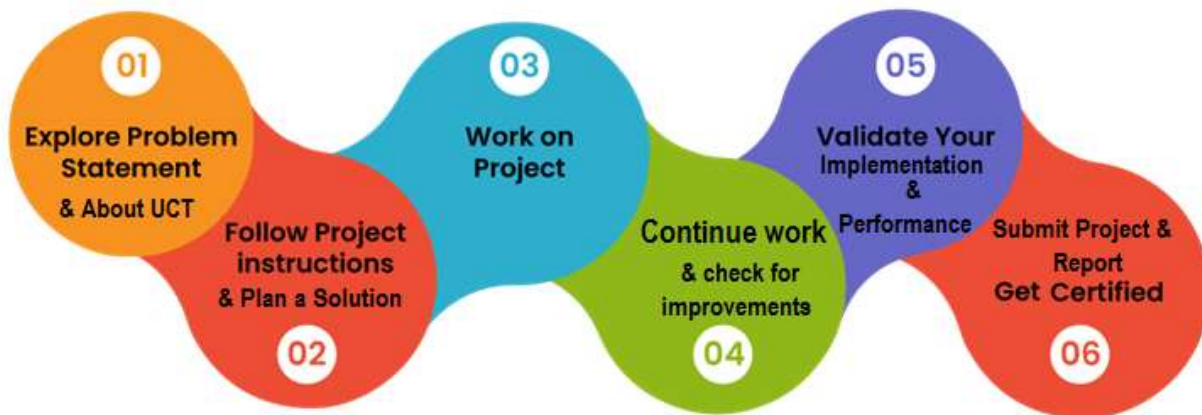
The six-week internship provided practical exposure to Python application development and software problem solving.

It helped me connect academic concepts with real implementation and improved my confidence in programming and debugging.

I developed four projects: URL Shortener, Secure Password Manager, Quiz Game and File Organizer.

The opportunity was provided through upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT).

The program followed a practical approach: understand requirements, select technologies, develop, test and document the projects.



I learned Python, Flask, SQLite, Tkinter, CustomTkinter, encryption and file handling, along with better testing and problem-solving practices.

I thank the internship coordinators, mentors, faculty members and everyone who supported me during the internship.

My message to juniors and peers is to focus on practical projects and improve them step by step.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.

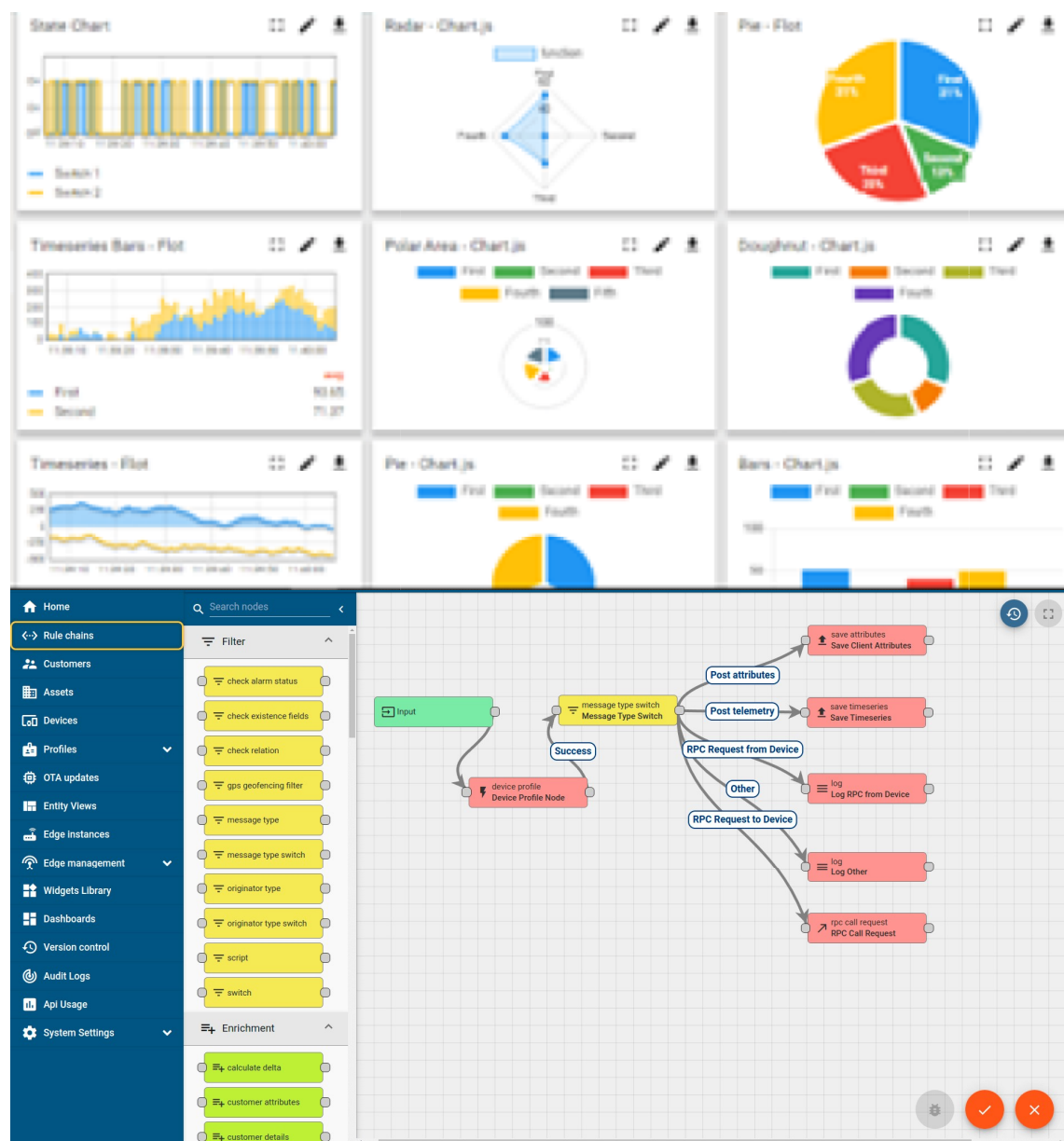


i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications while providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSQL Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to • Build Your own dashboard • Analytics and Reporting • Alert and Notification • Integration with third party application (Power BI, SAP, ERP) • Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (min)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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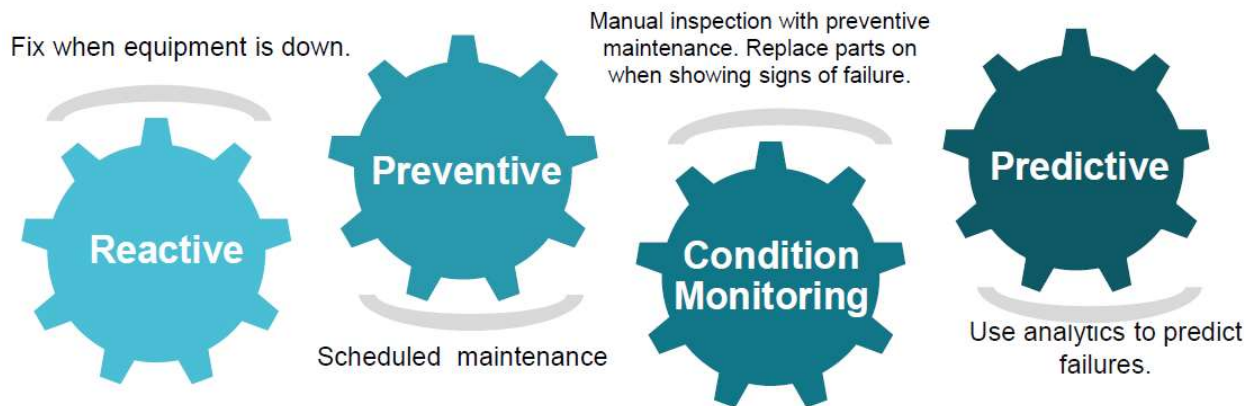


iii. LoRaWAN™ based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solutions in Agritech, Smart Cities, Industrial Monitoring, Smart Street Light, Smart Water/Gas/Electricity metering solutions etc.

iv. Predictive Maintenance

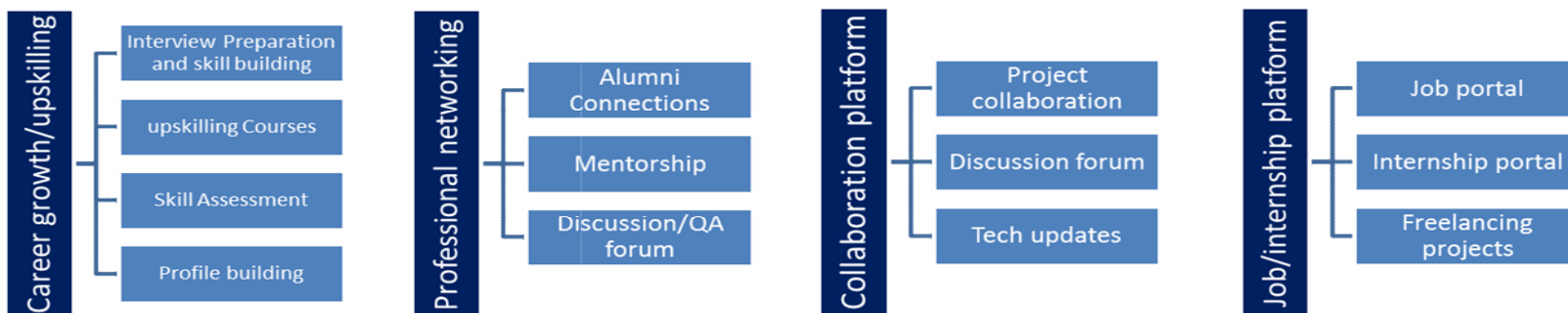
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded systems, Industrial IoT and Machine Learning Technologies by finding Remaining Useful Life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with UniConverge Technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT Academy is the EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- solve real-world problems.
- have improved job prospects.
- have improved understanding of our field and its applications.
- have personal growth like better communication and problem solving.

2.5 Reference

- [1] Upskill Campus / The IoT Academy Internship Material
- [2] UniConverge Technologies Pvt Ltd Internship Guidelines
- [3] Python documentation and project resources

2.6 Glossary

Terms	Acronym
GUI	Graphical User Interface
URL	Uniform Resource Locator
CRUD	Create, Read, Update and Delete
SQLite	Lightweight relational database
Flask	Python web framework

3 Problem Statement

The assigned problem statement was to develop practical Python applications that solve common software-use problems.

The four applications were a URL Shortener, Secure Password Manager, Quiz Game and File Organizer, each designed to demonstrate practical Python programming.

4 Existing and Proposed solution

Existing solutions often require manual work or separate tools. URLs, passwords, quiz results and files can all be managed manually, which is less convenient and less automated.

The proposed solution is a set of Python applications using Flask, SQLite, Tkinter, CustomTkinter, Fernet encryption and file-system operations.

The value addition is practical automation with simple interfaces and reusable programming concepts.

4.1 Code submission (Github link)

<https://github.com/Disha0708/upskill>

4.2 Report submission (Github link)

<https://github.com/Disha0708/upskill>

5 Proposed Design/ Model

The solutions follow an input-process-output flow. The URL Shortener maps URLs in SQLite; the Password Manager uses encrypted database storage; the Quiz Game processes answers and scores; and the File Organizer categorizes and moves files.

5.1 High Level Diagram (if applicable)¹

5.2 Low Level Diagram (if applicable)

Interfaces (if applicable)

URL Shortener: web interface. Password Manager: desktop GUI. Quiz Game: console interface. File Organizer: desktop GUI with folder selection and organize controls.

6 Performance Test

Performance testing verified that the applications work correctly as practical software solutions.

The main constraints were input handling, database reliability, password security, file categorization and score calculation.

These were handled through validation, SQLite queries, Fernet encryption, extension matching and answer comparison.

Testing was performed with normal, valid and invalid inputs and the results were compared with expected behavior.

Test cases covered URL generation/redirection, credential save/search/update/delete, correct and incorrect quiz answers, and file organization.

The results were satisfactory for the implemented features. Future testing can include scalability, stronger security and deployment-oriented checks.

- 6.1 Test Plan/ Test Cases: URL, password manager, quiz and file-organization workflows.**
- 6.2 Test Procedure: Execute each project, provide inputs, verify output and check database/file changes.**
- 6.3 Performance Outcome: Main functional workflows produced the expected results during testing.**

7 My learnings

The internship improved my Python programming, debugging and problem-solving skills. I gained practical exposure to Flask, SQLite, Tkinter, CustomTkinter, encryption, file handling and functional testing. These skills will support my future software-development work.

8 Future work scope

Future work can include URL analytics and custom aliases; stronger authentication and backup for the Password Manager; categories, timers and leaderboards for the Quiz Game; and scheduled organization, duplicate detection and undo functionality for the File Organizer.